PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS

39th ANNUAL CONVENTION

Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

6 OCTOBER 2026

Grand Ballroom, Okada Manila









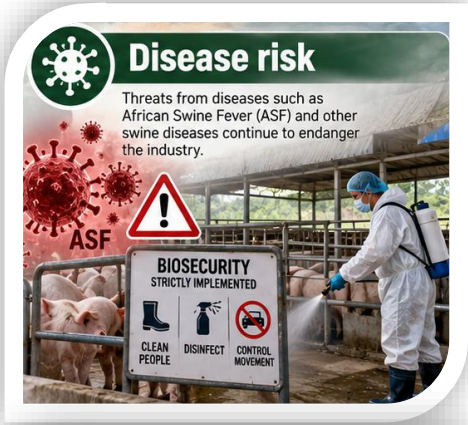

PHILIPPINE SOCIETY OF ANIMAL NUTRITIONISTS
39th Annual Convention | October 6, 2026

Feeding for Resilience: Industry Trends and Innovations in Swine Nutrition

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Swine Industry Status and Challenges



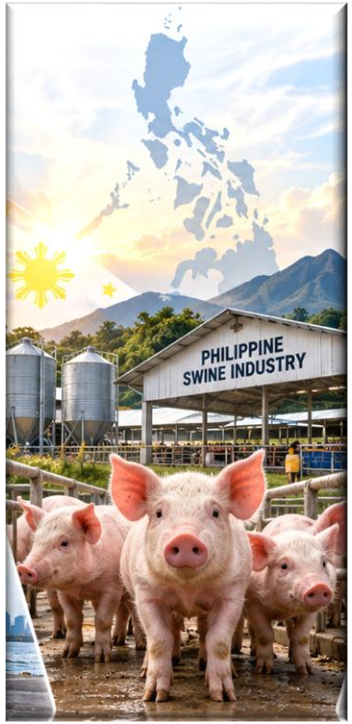






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The changing Philippine swine industry



- Road to recovery, but a different industry
- Not simply returning to the pre-ASF structure
- Emerging industry → larger, technically sophisticated commercial component alongside a still-large smallholder section
- Changes the objective of nutrition

"From least-cost formulation to resilient formulation"

Profitability + Robustness + Sustainability



Traditional KPIs
Old Production Model

- ADG
- FCR
- LSBA
- PWSY
- PSSY
- ₱/kg feed
- Production cost



Outcome-Based KPIs
Modern Production Model

- Full-value pigs marketed, %
- Whole-chain survivability, %
- kg pork sold/sow/year
- Feed/kg pork sold
- ₱ Feed cost/kg pork sold
- CV of market/carcass weight
- Sow lifetime kg pork produced

Technical developments are more relevant

Traditional formulation

- Static nutrient specifications
- Crude protein and total mineral targets
- Generous safety margins
- Ingredient averages used for long periods

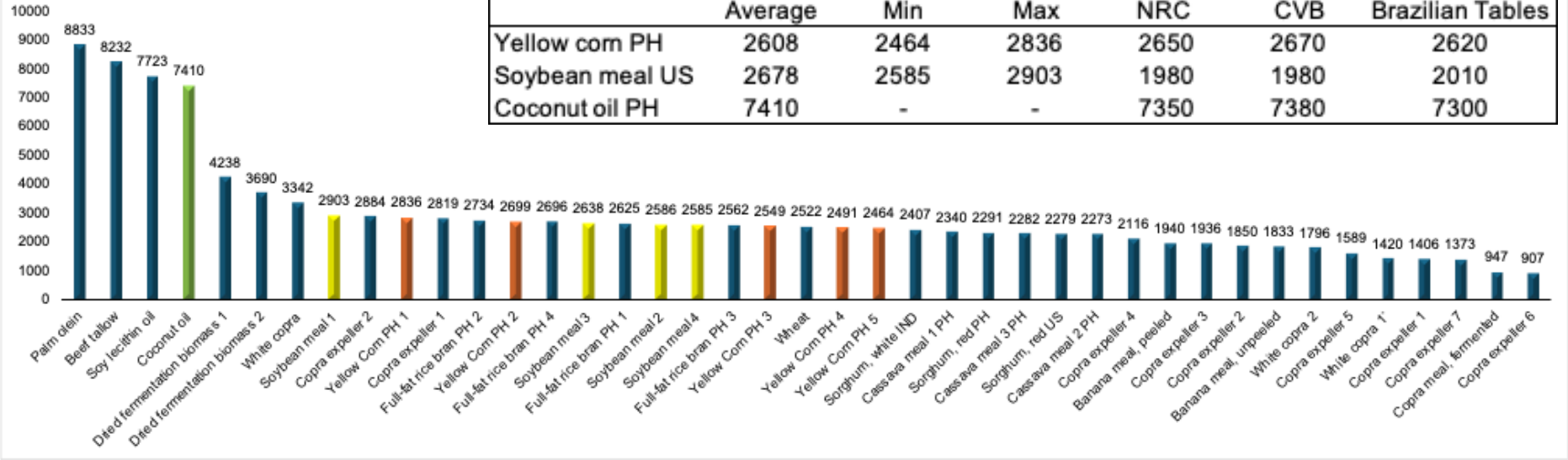
Precision formulation

- SID amino acids and digestible P/Ca
- ME⇒NE-based formulations
- Multi-phase feeding
- Near-real-time ingredient matrix updates

Swine NE of feed ingredients
(kcal/kg, as-fed basis, IAS UPLB)

n = 41 feed ingredients

	Average	Min	Max	NRC	CVB	Brazilian Tables
Yellow corn PH	2608	2464	2836	2650	2670	2620
Soybean meal US	2678	2585	2903	1980	1980	2010
Coconut oil PH	7410	-	-	7350	7380	7300



Sources: Abella et al. (2018), Adriano (2018), Arriesgado and Sulabo (2018), Bagui (2018), Dagandan and Sulabo (2019), de Castro et al. (2022), Espinosa and Sulabo (2015), Farma and Sulabo (2025), Maderse and Sulabo (2026), Manaig and Sulabo (2017), Ortiz (unpublished), Padua and Sulabo (2025), Villanueva and Sulabo (2018)

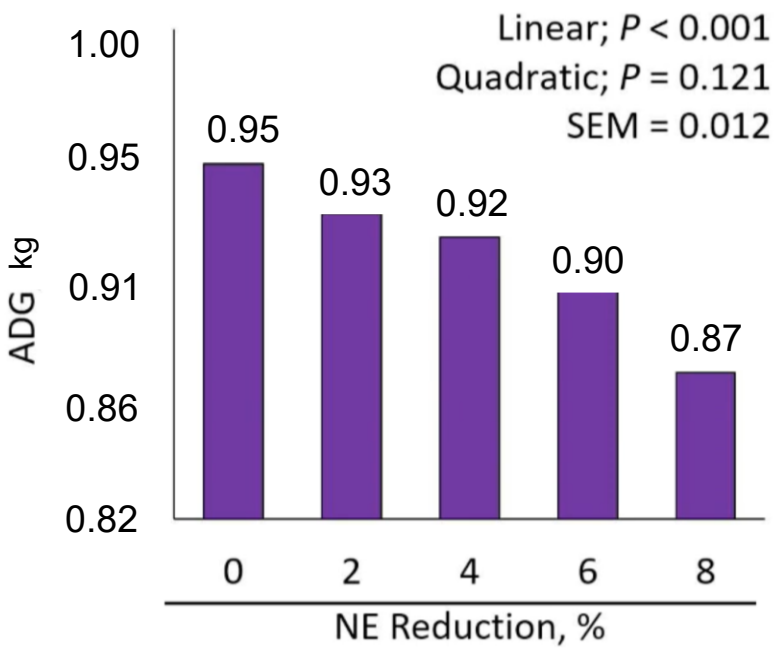
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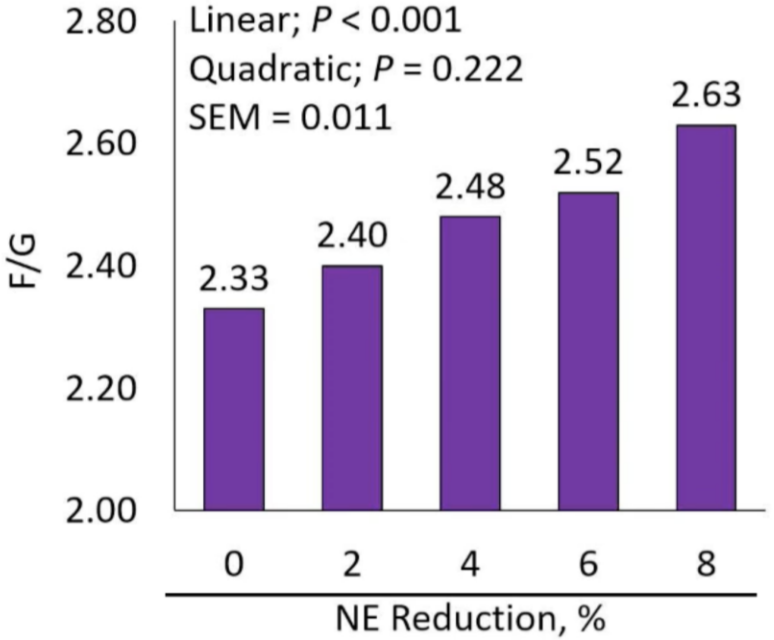
Phase 1 Diet Composition (as-fed basis)		
	NE Reduction, %	
Ingredient, %	0	8
Corn	64.32	31.77
Soybean meal	32.04	25.34
Wheat middlings	---	25.00
Corn DDGS	---	15.00
L-Lys HCl	0.45	0.34
Other AA	0.67	0.23
Vitamins and Minerals	2.52	2.35
Total	100.00	100.00

Calculated Analysis		
	NE Reduction, %	
SID AA, %	0	8
Lys, %	1.27	1.17
Ile:Lys	58	66
Leu:Lys	113	142
Met and Cys:Lys	59	59
Thr:Lys	66	66
Trp:Lys	20.0	20.1
Val:Lys	70	75
NE, kcal/kg	2,432	2,237
SID Lys:NE, g/Mcal	5.22	5.23
CP, %	20.5	23.1
Ca, %	0.70	0.79
STTD P, %	0.40	0.43

Royall et al. (2024)

Effect of Diet NE level on ADG and F/G

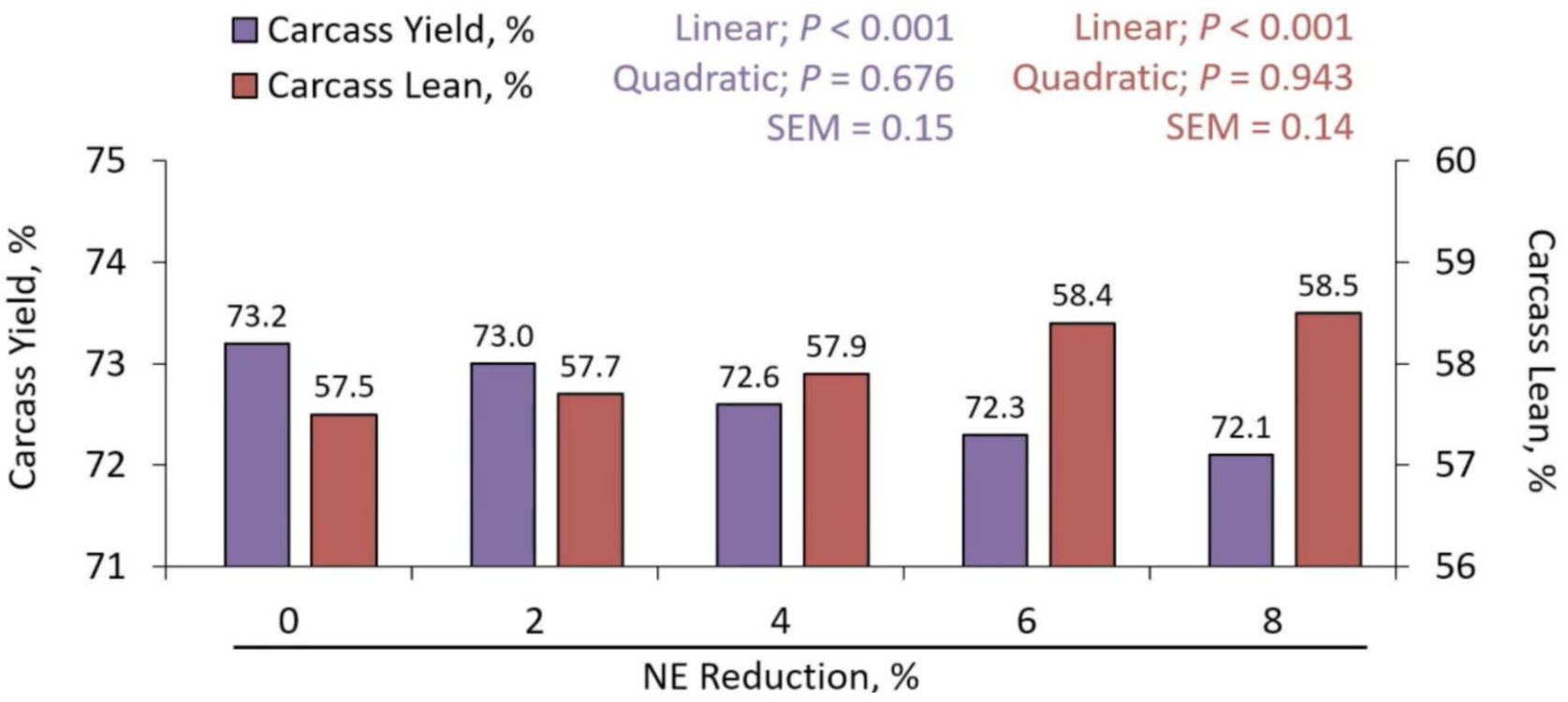




Royall et al. (2024)

/uplbcafsias

Effect of Diet NE level on Carcass Yield and Lean



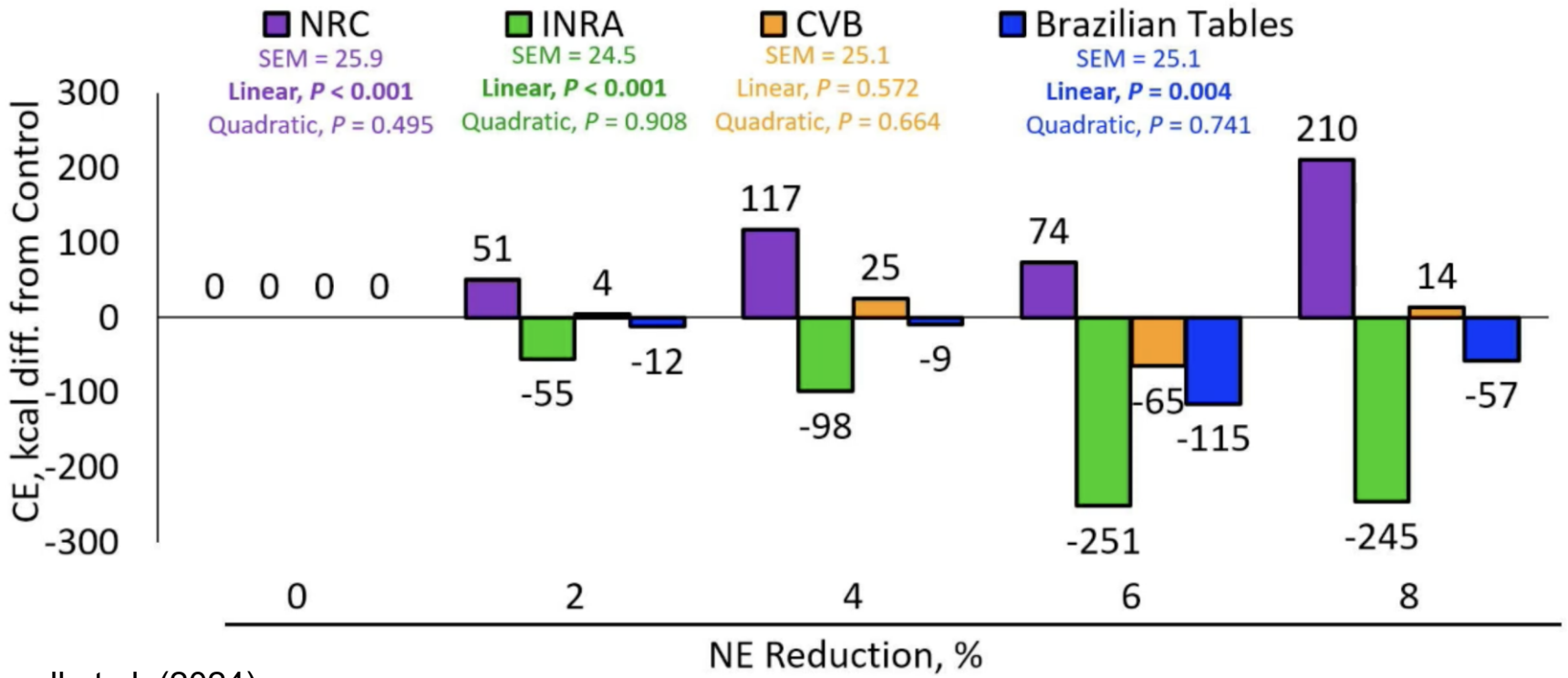
Royall et al. (2024)

Chemical Analysis of Major Ingredients				
Item, %	**Corn**	**SBM**	**Wheat Midds**	**Corn DDGS**
Dry Matter	85.2	87.6	86.6	89.2
Crude Protein	7.4	46.3	18.0	30.6
Starch	61.7	1.1	13.5	1.8
Ether Extract	3.1	1.8	3.6	6.0
Acid Detergent Fiber	2.7	4.8	12.3	11.2
Neutral Detergent Fiber	7.0	5.5	39.2	27.8
Ash	1.2	5.8	5.4	5.0

Ingredient Database NE, kcal/kg	Corn	SBM	Wheat Midds	Corn DDGS
NRC, 2012	2672	2088	2112 (79%)	2343 (88%)
INRA, 2007	2709	2110	1869 (69%)	2026 (75%)
CVB, 2020	2690	1975	1638 (61%)	22067 (82%)
Brazilian Tables, 2024	2665	2035	1693 (64%)	1971 (74%)

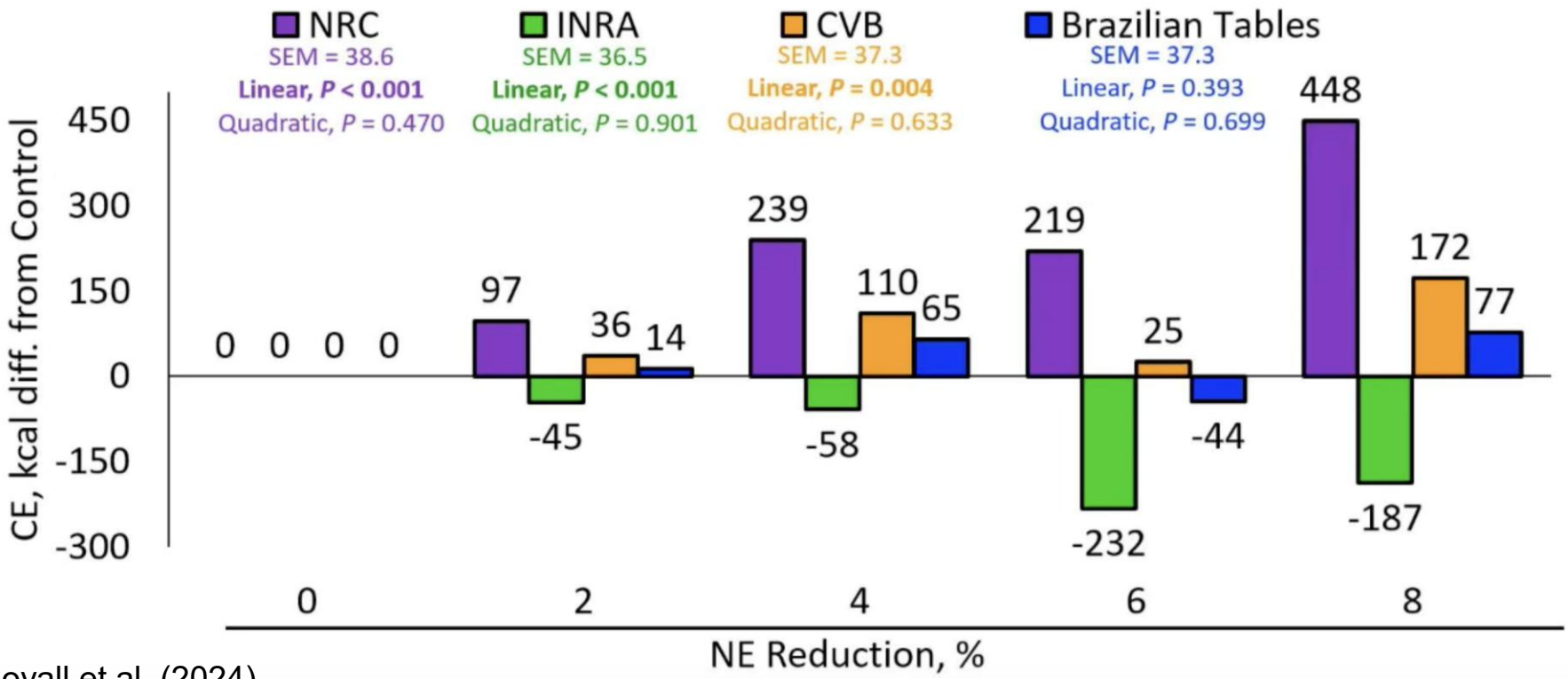
Royall et al. (2024)

Caloric Efficiency, Live Basis (Ingredient Databases)



Royall et al. (2024)

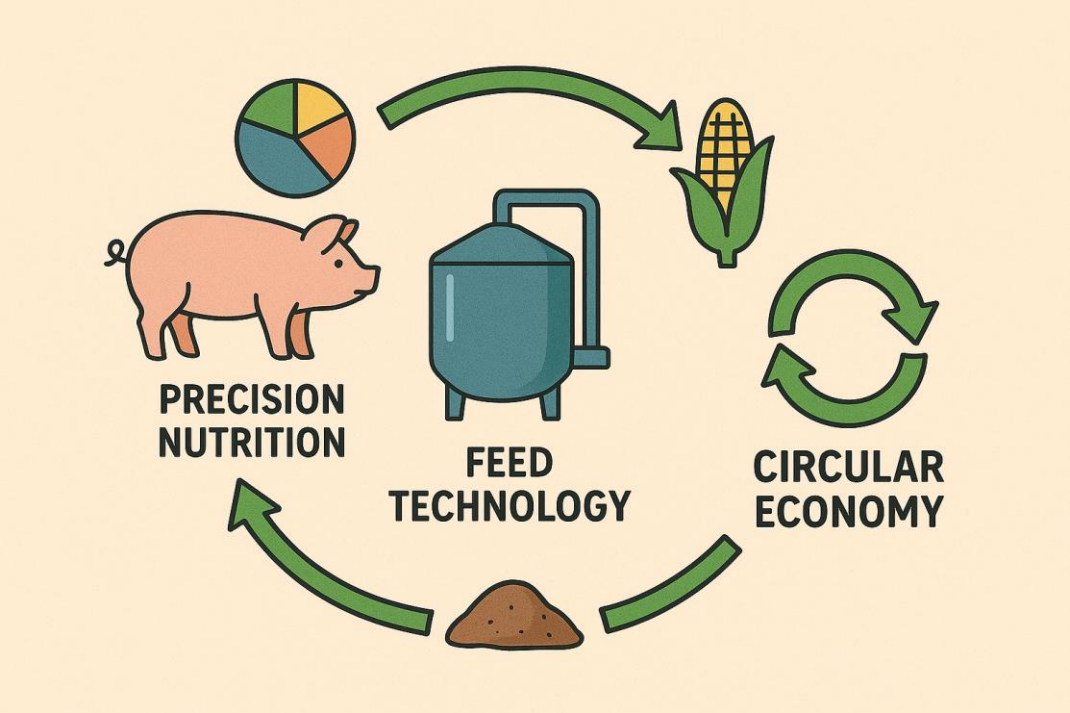
Caloric Efficiency, Carcass Basis (Ingredient Databases)



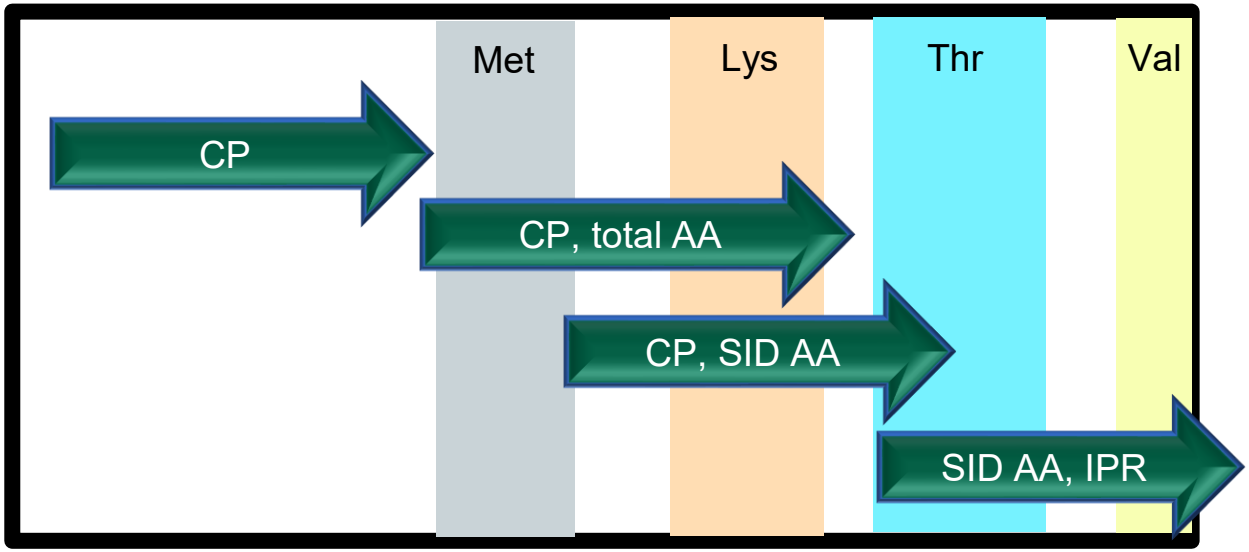
Royall et al. (2024)

Crude Protein Valorization

- Enhancing efficiency of dietary protein conversion



Refining precision of swine feed formulation



Establish optimal SID Lys level

Use appropriate ideal AA ratios

Eliminate CP restrictions

Effect of reducing CP in growing-finishing pig diets when formulated at constant SID Lys:NE ratios

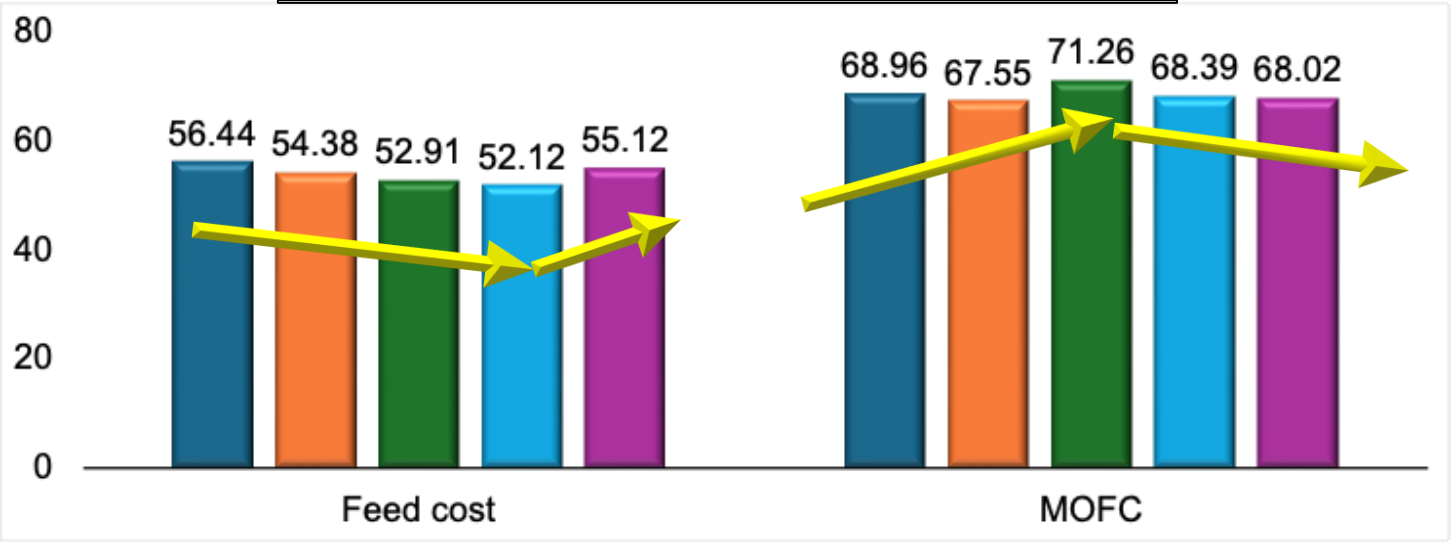
Item	Diet CP level, %					SEM	*P*-value
Grower:	21.00	19.99	17.97	16.32	14.78		
Finisher 1:	18.36	17.22	15.95	14.33	12.72		
Finisher 2:	16.93	15.56	15.17	13.11	11.81		
ADG, kg	0.906	0.881	0.897	0.873	0.888	0.02	0.57
ADFI, kg	2.775	2.772	2.821	2.811	2.776	0.04	0.84
F/G	3.070	3.155	3.161	3.227	3.135	0.07	0.63
Final wt, kg	108.8	106.6	107.0	106.7	107.5	1.33	0.55
BUN, mg/dL	17.9	14.9	15.7	12.0	10.6	1.03	<0.001
Backfat thickness (ave), mm	29.56	30.26	28.02	28.81	31.62	1.58	0.45
Standardized fat free lean, %	54.89	53.33	55.41	53.99	53.52	1.43	0.62
Marbling score	2.50	2.61	2.39	2.15	2.75	0.22	0.40

+L-Lys, L-Met, L-Thr, L-Val, L-Trp, L-Ile

Militante and Sulabo (2025)

Effect of reducing CP in growing-finishing pig diets when formulated at constant SID Lys:NE ratios

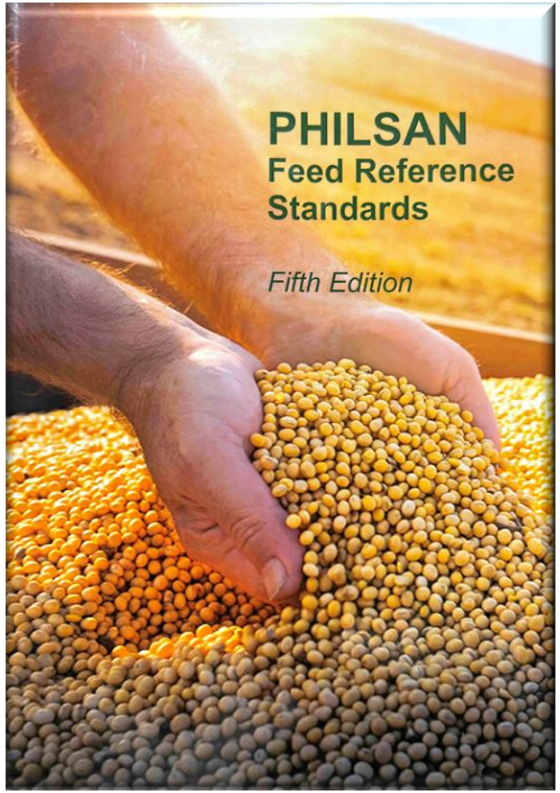
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Finisher 2:	16.93	15.56	15.17	13.11	11.81



Militante and Sulabo (2025)

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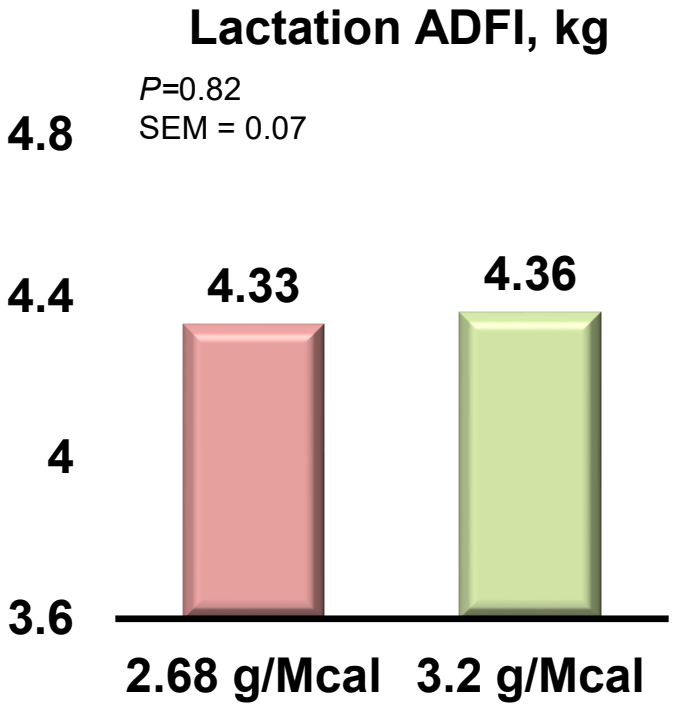
PHILSAN Swine Nutrient Recommendations (2026)

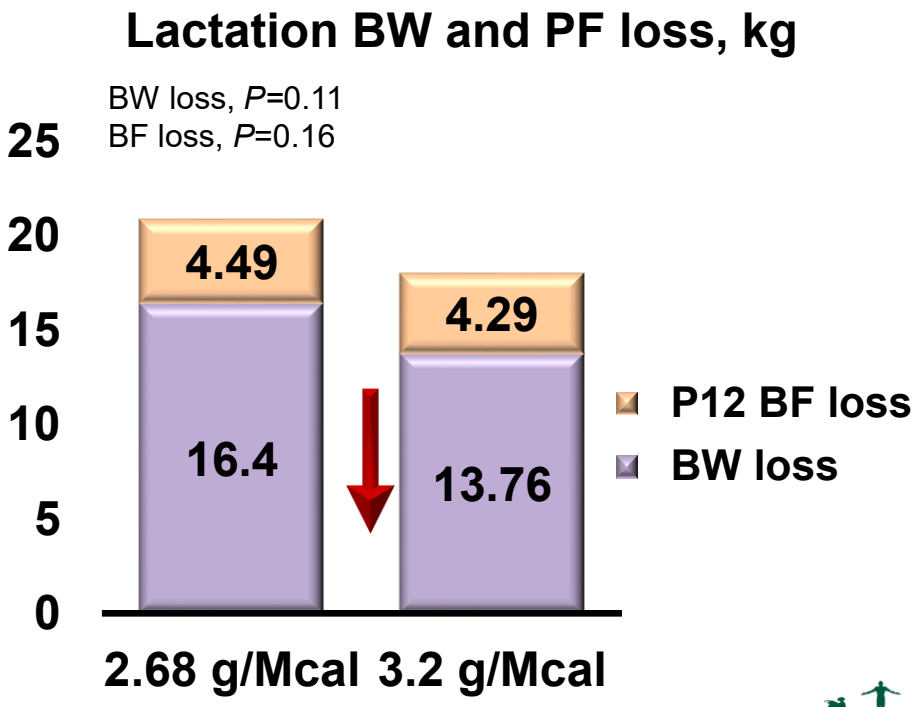




Recommendations derived using defined assumptions for feed intake and reproductive performance ➜ modern breeders need dynamic nutrient supply

Effect of SID Lys:calorie ratio on sow performance



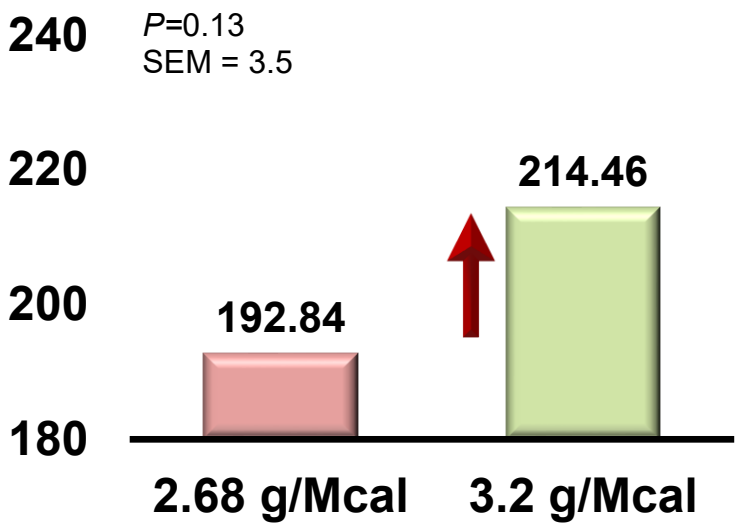


Balot and Sulabo (2025)

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Effect of SID Lys:calorie ratio on progeny performance

Piglet ADG, g



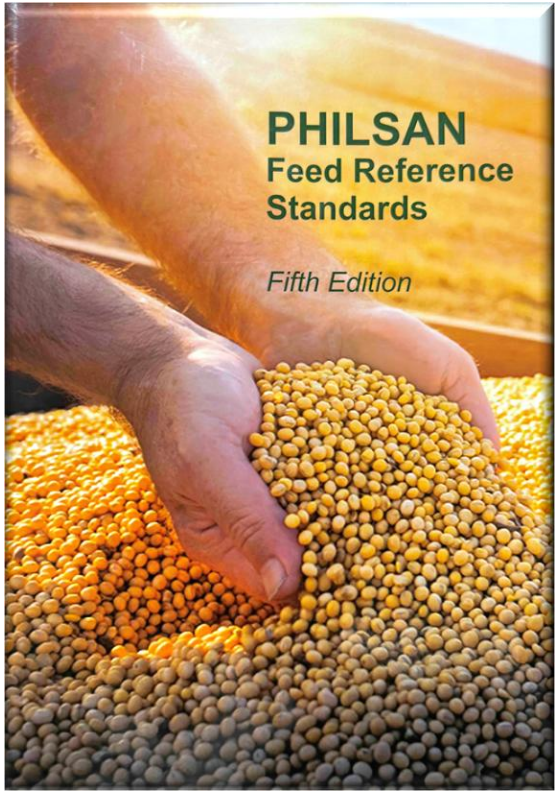
Sow milk composition

Item	SID Lys:ME			
	2.68 g/Mcal	3.2 g/Mcal	SEM	*P*-value
Milk yield, g	588.39	642.84	9.17	0.13
DM, g	113.42	122.16	1.31	0.13
Energy, kcal	747.11	802.42	8.31	0.13
Nitrogen, g	27.42	30.24	0.44	0.14

Balot and Sulabo (2025)

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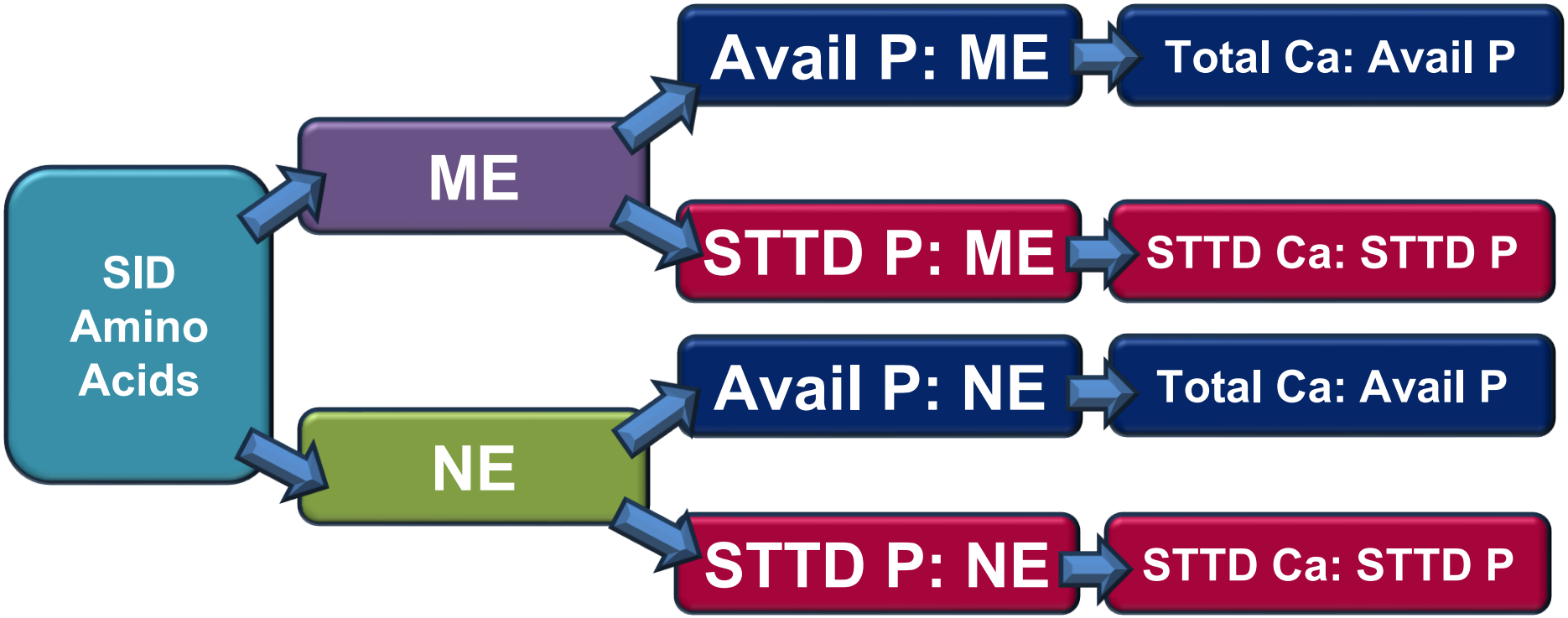
PHILSAN Swine Nutrient Recommendations (2026)



High Lean Gain	PDR >140 g/d Lean Gain >363 g/d ADG >900 g/d
Medium Lean Gain	PDR 115-140 g/d Lean Gain 295-363 g/d ADG 800-900 g/d
Low Lean Gain	PDR <115 g/d Lean Gain <295 g/d ADG <800 g/d

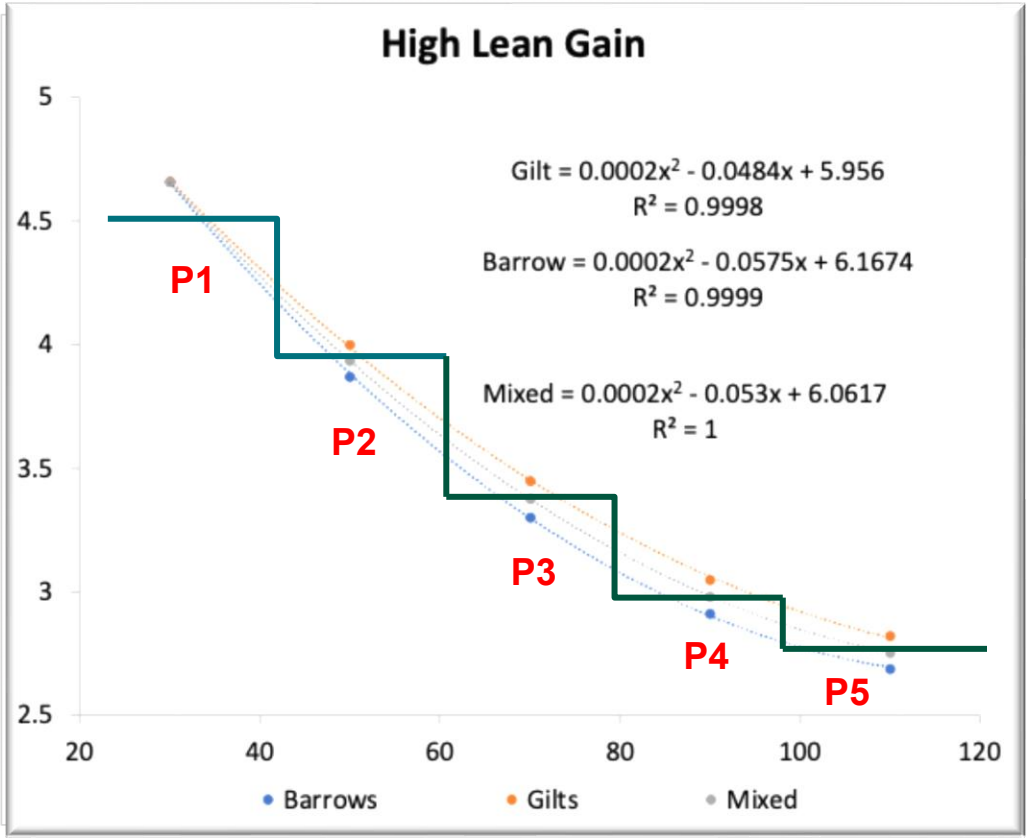
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PHILSAN Swine Nutrient Recommendations (2026)



SID Lys:NE models

	SID Lys:NE, g/Mcal	
Weight, kg	Gilts	Barrows
20	5.14	5.03
30	4.72	4.60
40	4.34	4.21
50	4.00	3.87
60	3.70	3.56
70	3.45	3.30
80	3.23	3.09
90	3.05	2.91
100	2.92	2.78
110	2.82	2.69
120	2.76	2.64
130	2.75	2.63



PHILSAN (2026)

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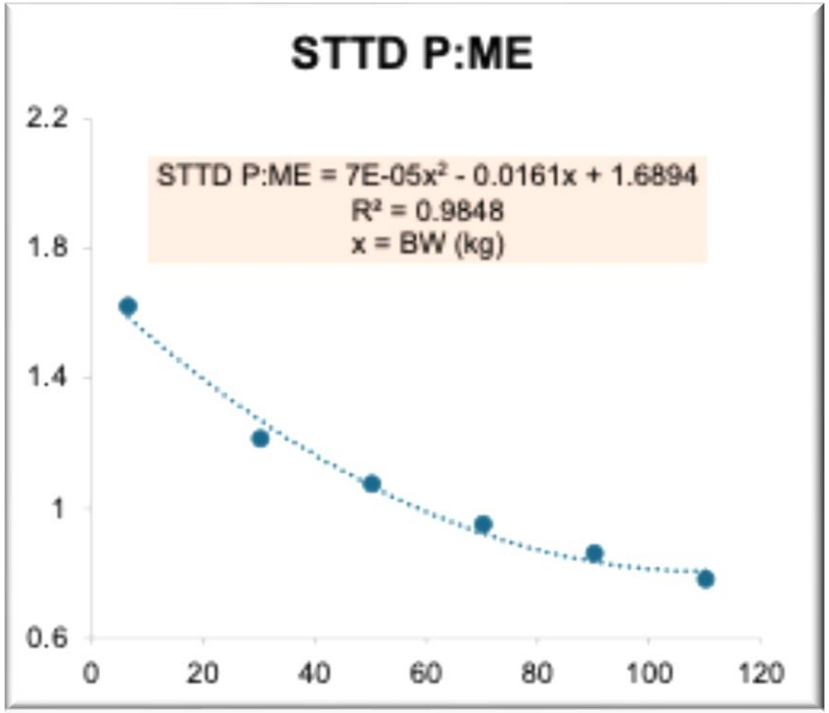
Recommended ideal protein ratios

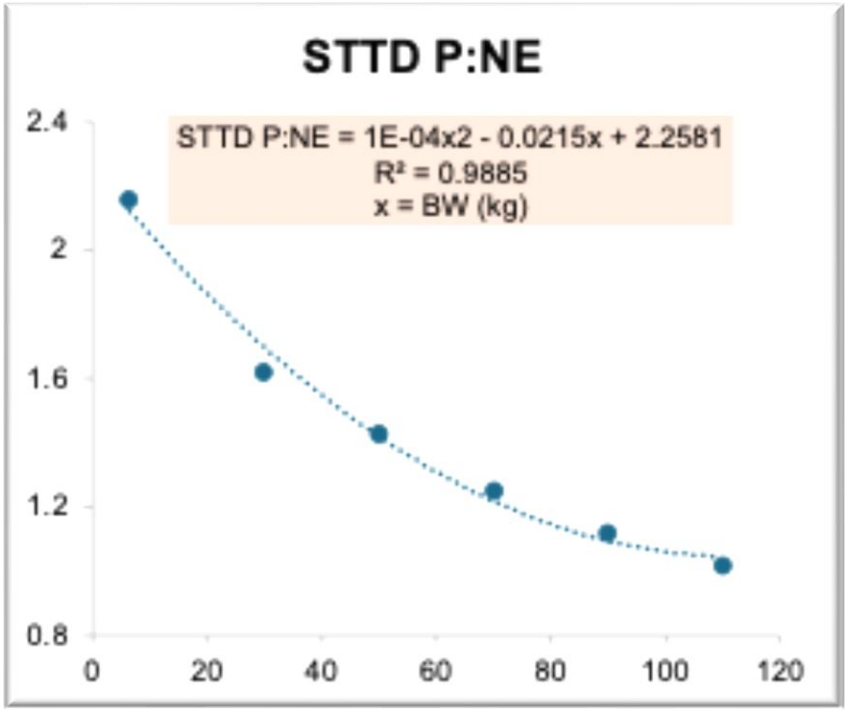
(Growing-finishing pigs)

Body weight	20 to 60 kg			60 to 100 kg			> 100 kg		
Item	Rec.	Range		Rec.	Range		Rec.	Range	
Lysine	100	--	--	100	--	--	100	--	--
Threonine	63	60	67	65	60	68	67	60	72
Methionine	32	31	33	33	31	34	34	31	36
Methionine+Cysteine	58	56	60	60	57	61	62	57	65
Tryptophan	18	15	20	18	15	20	18	15	20
Isoleucine	55	51	57	55	52	58	55	52	60
Valine	67	64	70	67	64	69	67	64	74
Arginine	46	46	46	46	46	46	46	46	46
Histidine	33	32	36	33	32	36	33	32	35
Leucine	100	100	108	100	100	108	100	100	102
Phenylalanine	60	60	60	60	60	60	60	60	60
Phenylalanine+Tyrosine	95	94	113	95	94	113	95	94	102

PHILSAN (2026)

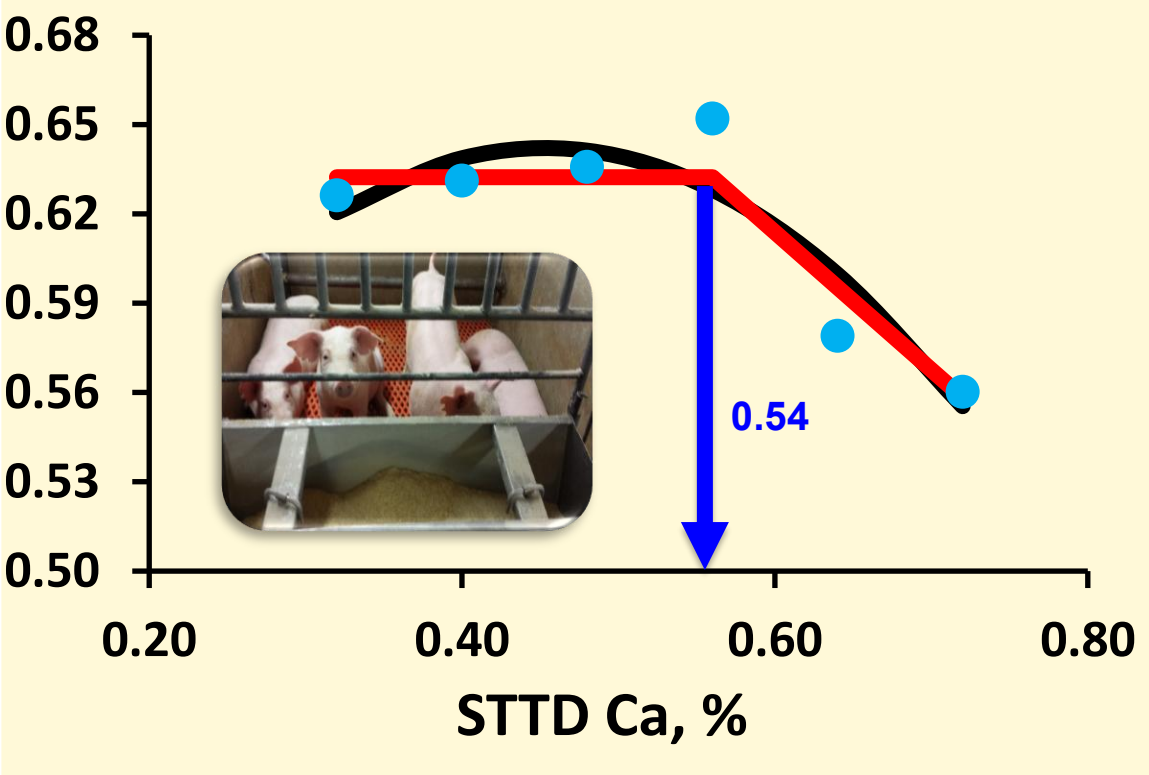
STTD P requirement in pigs from 5 to 120 kg





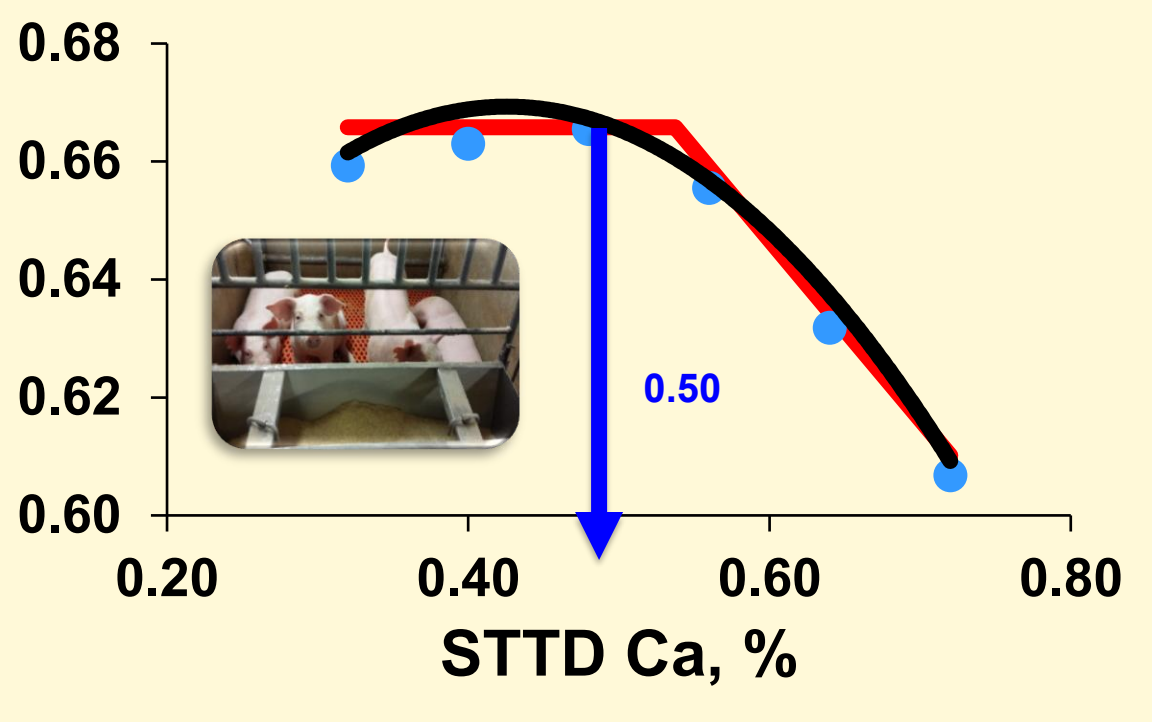
PHILSAN (2026)

Effect of diet Ca level on ADG of 11-25 kg pigs



Gonzalez-Vega et al. (2016a)

Effect of diet Ca level on G:F of 11-25 kg pigs



Gonzalez-Vega et al. (2016a)

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High Ca

Low Ca





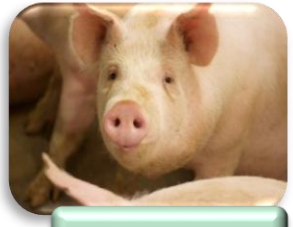
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Requirements for STTD Ca:STTD P in growing pigs









STTD P, %	0.43	0.39	0.30	0.25
Growth performance	1.40:1	1.35:1	1.25:1	1.10:1
STTD Ca, %	0.60	0.53	0.38	0.28

[1]PHILSAN (2026)
[2]González-Vega et al. (2016a); Merriman et al. (2017); Lagos et al. (2019a); Lagos et al. (2019b)

Feeding program design, example
(Growing-finishing pigs)

	Starter	Grower	Finisher 1	Finisher 2
Body weight, kg	20 to 40	40 to 60	60 to 80	80 to 100
Sex	Mixed	Mixed	Mixed	Mixed
Assumed ADFI, kg	1.40	2.12	2.70	3.14
NE intake, Mcal/d	3.36	5.09	6.48	7.54
Diet NE, Mcal/kg	**2.40**	**2.40**	**2.40**	**2.40**
SID Lys:NE, g/Mcal	**4.66**	**3.94**	**3.38**	**2.98**

Step 2.

Step 1.

Determine appropriate SID Lys:NE ratio according to feeding phase

Feeding program design, example
(Growing-finishing pigs)

	Starter	Grower	Finisher 1	Finisher 2
Body weight, kg	20 to 40	40 to 60	60 to 80	80 to 100
SID Lys:NE, g/Mcal	**4.66**	**3.94**	**3.38**	**2.98**
Diet NE, Mcal/kg	2.40	2.40	2.40	2.40
SID AA, %				
Lysine	**1.12**	**0.95**	**0.81**	**0.72**
Threonine	0.70	0.60	0.53	0.46
Methionine	0.36	0.30	0.27	0.24
Methionine+Cysteine	0.65	0.55	0.49	0.43
Tryptophan	0.20	0.17	0.15	0.13
Isoleucine	0.62	0.52	0.45	0.39
Valine	0.75	0.63	0.54	0.48
Arginine	0.51	0.43	0.37	0.33
Histidine	0.37	0.31	0.27	0.24
Leucine	1.12	0.95	0.81	0.72
Phenylalanine	0.67	0.57	0.49	0.43
Phenylalanine+Tyrosine	1.06	0.90	0.77	0.68

Step 3.

Calculate: (4.66 x 2.4)/10

Step 4.

Calculate the rest of AA using AA:Lys ratios

/uplbcafsias

Feeding program design, example
(Growing-finishing pigs)

	Starter	Grower	Finisher 1	Finisher 2
Body weight, kg	20 to 40	40 to 60	60 to 80	80 to 100
Sex	Mixed	Mixed	Mixed	Mixed
Assumed ADFI, kg	1.40	2.12	2.70	3.14
Diet NE, Mcal/kg	2.40	2.40	2.40	2.40
STTD P:NE, g/Mcal	1.62	1.43	1.25	1.12
STTD P, % of diet	0.39	0.34	0.30	0.27
STTD P, g/d	5.45	7.26	8.11	8.41
STTD Ca, % of diet	0.53	0.44	0.38	0.32
STTD Ca, g/d	7.35	9.43	10.14	10.09

Step 5.

Step 6

Calculate the digestible P:energy ratio

Calculate the digestible P, and then calculate the Ca level using the ideal Ca:P ratio

Takeaway Messages

Feed for resilience, not simply for least cost.



Build resilience through precision nutrition.



Design feeding programs around nutrient efficiency and flexibility.



Measure resilience by whole-system outcomes.





Thank You Very Much!